

Reserve review: US commercial auto (NAIC 1767) as at 31 December 2007

Illustrative reserving study on NAIC Schedule P data

Data: CAS Loss Reserving Database (NAIC Schedule P), compiled by S&P Global Market Intelligence for the Casualty Actuarial Society. Net of reinsurance, US business. The IFRS 17 and Solvency II figures are illustrative methodology demonstrations, not accounting or regulatory numbers for any company. Insurers are identified by NAIC code only. Amounts in USD millions unless stated.

1. Executive reserve review

Selected unpaid claims at 31 December 2007 are USD 365.4m, net of reinsurance, for accident years 1998–2007. This is USD 112.3m (44%) above the insurer’s own booked unpaid of USD 253.1m. The estimate covers commercial auto losses including defence and cost containment (DCC), undiscounted and excluding ULAE. It was locked in git (tag `selection-locked`) before any later development was looked at.

	USD m
Selected ultimate, AY 1998–2007	1876.9
Paid to date	1511.5
Selected unpaid	365.4
of which case reserves	117.0
of which IBNR (incl. IBNER)	248.4
Insurer’s booked unpaid (case + bulk/IBNR)	253.1
Range across methods (paid CL to incurred BF)	343.5 – 370.2

Main drivers.

- AY 2006–2007 account for USD 255.4m (70%) of the unpaid. They are 58% and 33% paid-developed, and are selected with Benktander at loss ratios of 61.9% and 63.8%.
- The insurer books AY 2007 at a 49.0% loss ratio and AY 2006 at 53.7%. That implies much faster payment than its own history supports: paid chain ladder gives 61.4% for AY 2007.
- The insurer’s own first bookings have always developed upwards: every AY 1998–2006 booked ultimate has risen since its first valuation, by 8% to 30%. This pre-lock evidence supports holding more than the booked figure.
- The tail beyond 120 months is 1.0023 (log-linear extrapolation). It is immaterial, at under 0.3% of ultimate.

2. Evidence and uncertainty

Data and diagnostics. The paid and case-incurred triangles are complete and clean: no negative paid increments and no negative case reserves. The paid ÷ incurred ratio is stable by accident year, so neither faster settlement nor a change in case-reserve strength is evident. Mid-lag paid factors have risen for recent accident years (the 3–4 factor averages 1.13 for AY 1998–2001 and 1.16–1.18 for AY 2002–2004), and incurred factors show the same rise. `cyEffTest` does not reject (paid $Z = 12$), while `dfCorTest` rejects independence for paid, so Mack standard errors are treated as a benchmark only. Factors are volume-weighted over the latest five diagonals.

Pseudo-historical check (pre-lock). The rules were rerun at 2004, 2005 and 2006 and compared with paid diagonals up to 2007. Paid chain ladder under-predicted payments every time (A/E 1.15, 1.20, 1.09). The default rule did better (1.08, 1.14, 1.07). Using incurred chain ladder for mature years brought the A/E closer to 1 (`outputs/m4_pseudo_historical_by_ay.csv`). The latest-5 and all-year windows gave the same A/E to within 0.5%.

Selection by accident year. Default rule on paid percent developed z : chain ladder at $z \geq 70\%$, Benktander at 30–70%, and BF below 30%. BF uses a leave-one-out Cape Cod prior, 65.3%–67.2%. One documented exception applies: for mature years (AY 1998–2005, $z \geq 70\%$), incurred chain ladder replaces paid chain ladder, based on the pseudo-historical evidence above. No accident year shows a paid vs incurred chain ladder gap above 10%. The prior is more than 10 points above

the peer median loss ratio for recent years, but the peer panel is mostly small insurers with a different mix, so the prior was kept and its ± 5 -point sensitivity is part of the uncertainty analysis.

Table 1: Ultimates by method and selection, USD m (z = paid percent developed; CL_INC = incurred chain ladder; BK = Benktander).

AY	Prem	Paid	z	Paid CL	Inc CL	Paid BF	Benkt.	Method	Selected	ULR	Unpaid
1998	245.0	158.0	100%	158.4	158.4	158.4	158.4	CL_INC	158.4	64.6%	0.4
1999	231.5	158.4	100%	159.2	159.1	159.1	159.2	CL_INC	159.1	68.7%	0.7
2000	222.2	170.7	99%	172.7	175.8	172.4	172.7	CL_INC	175.8	79.1%	5.1
2001	233.6	149.9	98%	153.1	152.4	153.1	153.1	CL_INC	152.4	65.3%	2.5
2002	259.3	173.8	97%	180.0	180.3	179.7	180.0	CL_INC	180.3	69.5%	6.5
2003	281.5	173.3	94%	185.0	187.1	185.1	185.0	CL_INC	187.1	66.5%	13.8
2004	301.6	159.2	87%	182.3	182.8	184.9	182.6	CL_INC	182.8	60.6%	23.7
2005	322.8	167.3	76%	220.9	224.6	219.2	220.5	CL_INC	224.6	69.6%	57.3
2006	354.9	126.1	58%	216.0	218.9	225.1	219.8	BK	219.8	61.9%	93.7
2007	370.6	74.7	33%	227.6	227.5	240.9	236.5	BK	236.5	63.8%	161.8

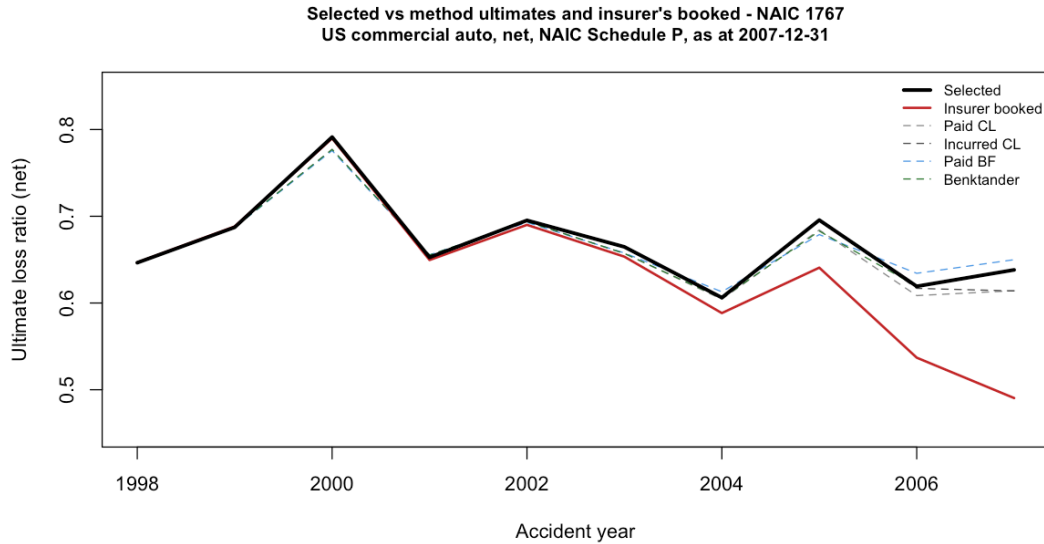


Figure 1: Selected vs method ultimates and the insurer's booked ultimate (loss ratio). US commercial auto, net, NAIC Schedule P, as at 2007-12-31.

Uncertainty. *To be completed after the lock with the frozen stochastic specification (M6): Mack and ODP bootstrap standard errors, the custom bootstrap of the selected reserve, and sensitivities (ELR ± 5 points, tail, window, incurred vs paid, calendar-year inflation, thresholds).*

3. Validation and limitations (post-reveal addendum)

Written after the reveal (M5).